

Crime Sense : Video Based Crime Classification using Contrast-Enhanced Residual-BiLSTM

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Abstract

In today’s cities, thousands of hours of surveillance video stream in uninterrupted—far more than any human team can watch. Conventional anomaly detectors can raise an alarm when “something unusual” occurs, but they leave operators guessing whether it was a robbery, a fight, or simply a stray animal. To close this gap, we present CrimeSense, an end-to-end pipeline that not only flags abnormal events but also identifies their type across fourteen real-world categories—ranging from Abuse and Burglary to Road Accidents and Vandalism—using the challenging UCF-Crime dataset (1,900 clips, 30 s–10 min each, captured under diverse, uncontrolled conditions).

Proposed method segments each video into 16-frame clips, resizes frames to 64×64 pixels, and applies Contrast-Limited Adaptive Histogram Equalization (CLAHE) to enhance local contrast, bringing out subtle edges and textures. These pre-processed clips feed into a lightweight residual encoder wrapped in a TimeDistributed layer: identity skip connections guarantee smooth gradient flow, while SpatialDropout1D randomly silences entire feature channels—simulating occlusions and boosting robustness. A bidirectional LSTM then reads the resulting 16-step feature sequence both forward and backward, capturing not only the lead-in motions but also the critical after-effects of suspicious behavior. The entire network—just 1.2 million parameters in total—trains and runs comfortably on a single GPU.

We benchmark two tasks: a binary Normal vs. Crime classifier (900 training clips per class) and a four-way Crimes categorization (Violent Crimes, Property Theft, Destructive Crimes, Legal/Accidental Incidents—200 clips per group). Using balanced train/validation/test splits and on-the-fly augmentations (random flips, brightness/contrast jitter, Gaussian blur), proposed models achieve 94.0% accuracy on the binary task and 95.62% on the four-way classification. These results outstrip heavier 3D-CNNs and frame-wise CNN + LSTM baselines. Finally, we demonstrate real-time usability by packaging CrimeSense in a Flask web service: users upload an MP4, and within seconds receive a “Crime / Normal” verdict followed by a detailed crime type.

By uniting adaptive contrast enhancement, efficient residual encoding, structured dropout, and bidirectional temporal modeling, CrimeSense offers a practical, high-fidelity framework for automated crime classification at city-scale.

Keywords: Video Based CrimeClassification, Anomaly Detection, Convolutional Neural Networks, Contrast-Limited Adaptive Histogram Equalization (CLAHE), Residual Convolutional Neural Network, Long Short Term Memory, Video-Based Analysis, Deep Learning

1 Introduction

In the past decade, the deployment of closed-circuit television (CCTV) and IP-based surveillance cameras has proliferated across urban and rural landscapes alike. From city streets and public transport hubs to retail stores and office buildings, cameras continuously record streams of video, generating petabytes of data every day. Paradoxically, this abundance of footage has created a “needle in a haystack” problem: while the raw volume of surveillance content has grown exponentially, human operators—tasked with monitoring screens for suspicious activity—face increasing fatigue, tunnel vision, and missed events [1]. A single security analyst cannot realistically watch hundreds of live feeds or sift through hours of recorded video without error.

Public- and private-sector surveillance systems have evolved from niche security installations into sprawling networks that blanket our cities. Gone are the days when a handful of fixed CCTV cameras sufficed; today’s surveillance architectures include high-definition PTZ units, wide-angle lenses, thermal imagers, and even mobile drones, all streaming video to central servers. This diversity of hardware delivers comprehensive coverage—but it also introduces a host of practical and technical challenges that any automated classification system must confront head-on.

To address this scalability challenge, automated video analytics systems have emerged, leveraging advances in computer vision and machine learning to assist—or even replace—manual monitoring [2]. Early systems focused on motion detection and simple object tracking, raising rudimentary alarms whenever unexpected movement occurred in a pre-defined region of interest. More sophisticated approaches adopted anomaly detection, modeling “normal” patterns of scene activity (pedestrian flow, vehicular traffic) and flagging statistically rare deviations. While such methods can alert operators to “something unusual,” they offer little insight into the nature of the event: Was it an altercation between individuals? A collision between vehicles? A case of theft or vandalism?

The distinction between anomaly detection and event classification is critical in real-world security contexts. An alarm that merely signals “anomaly detected” requires human review to determine whether the anomaly constitutes a security threat or benign variation (e.g., a windblown trash can crossing a sidewalk). In contrast, automated classification into semantically meaningful categories—such as “violent assault,” “shoplifting,” “road accident,” or “arson”—empowers rapid, context-aware responses. Law enforcement can prioritize dispatch to violent incidents over minor

disturbances; store managers can identify and review theft footage without wading through irrelevant clips; emergency services can allocate resources to traffic collisions with greater urgency [3]. Thus, developing models that not only detect but also accurately classify specific crime types in untrimmed surveillance video represents a pivotal advancement in intelligent video analytics.

However, video-based crime classification poses several formidable technical challenges. First, surveillance video is typically untrimmed, spanning many minutes or hours with anomalies occupying only a few seconds or less. Extracting informative snippets necessitates efficient sampling or sliding-window strategies that capture the temporal span of events without overwhelming computational budgets. Second, the video data often suffer from low resolution, poor lighting, and significant sensor noise conditions under which standard convolutional neural networks (CNNs) trained on high-quality, well-lit datasets struggle to generalize. Third, anomalies manifest not only in spatial patterns (object shapes, scene changes) but also in temporal dynamics: the buildup of suspicious behavior, the abrupt onset of harmful actions, and the subsequent aftermath. Models must therefore integrate robust spatial feature extraction with temporal sequence modeling to fully characterize crime events[4].

The research community has made commendable progress in video anomaly detection, employing techniques ranging from autoencoders that reconstruct “normal” video sequences to one-class support vector machines on handcrafted features [5]. Yet the extension from binary detection to multiclass crime classification remains relatively underexplored. Public benchmarks such as UCF-Crime, ShanghaiTech, and UCSD Pedestrian provide valuable data for anomaly detection, but datasets specifically annotated for fine-grained crime categories are scarce and often limited in scale. Moreover, many state-of-the-art video classification architectures—3D CNNs such as C3D and I3D, or two-stream networks combining RGB and optical flow—carry tens of millions of parameters, demanding extensive compute resources and large-scale annotated corpora to train effectively [6]. For practical deployment at the edge or in resource-constrained security centers, lightweight yet accurate models are essential.

This thesis addresses these gaps by proposing a unified, end-to-end pipeline for video-based crime classification that balances spatial depth, temporal expressiveness, and computational efficiency. The Proposed Approach begins with a careful frame sampling strategy, extracting short, fixed length clips that capture the full temporal extent of anomalies within long, untrimmed videos. We then apply a targeted, frame-level contrast enhancement to mitigate the effects of poor lighting and sensor noise, ensuring that subtle spatial cues—such as the flash of a weapon or the silhouette of a perpetrator are brought into relief [7]. At the heart of the model lies a deep residual convolutional backbone, designed to learn robust spatial features without the vanishing-gradient or overfitting issues that plague very deep networks on small datasets. We complement this backbone with a bidirectional recurrent network that synthesizes both the build-up and aftermath of crime events, capturing the full context needed for accurate classification [8]. To guard against overfitting and enforce robustness to missing spatial cues, we integrate structured regularization in the form of spatial dropout, selectively removing entire feature channels during training.

Through extensive experiments on the challenging Anomaly detection dataset UCF comprising nearly two thousand real world surveillance videos annotated for thirteen distinct anomaly types plus normal activity we demonstrate that this hybrid CNN BiLSTM pipeline achieves state-of-the-art performance. On the binary Normal vs. Crime task, the proposed lightweight model surpasses heavier pretrained networks, delivering over 90% precision and recall while maintaining a footprint of only 1.2 million parameters. In the more demanding four-way crime classification scenario, grouping anomalies into semantically coherent categories such as “Violent Crime,” “Property Theft,” “Destructive Crime,” and “Legal/Accidental Incidents,” the proposed approach maintains high accuracy, outpacing baseline methods by substantial margins. These results underscore the feasibility of deploying sophisticated video classifiers in real-time or near-real-time settings, an imperative for modern surveillance applications.

In this study, we present a streamlined yet powerful pipeline for automated crime event classification in real-world surveillance videos. Beginning with the UCF–Crime dataset’s 1,900 untrimmed clips, we segment each video into fixed sixteen-frame snippets, apply Contrast-Limited Adaptive Histogram Equalization (CLAHE) to enhance local contrast under challenging lighting, and resize frames to a computationally efficient 64×64 resolution. These preprocessed clips feed into a lightweight, residual convolutional encoder wrapped in a TimeDistributed block, where identity skip-connections and SpatialDropout1D strengthen gradient flow and robustness against occlusions. A bidirectional LSTM then ingests the resulting sequence of per-frame embeddings, capturing both “lead-in” and “follow-through” temporal cues that distinguish subtle crime dynamics. Finally, a compact classification head—with a single ReLU-activated dense layer and softmax output—seamlessly switches between binary (Normal vs. Crime) and four-way (Violent, Property Theft, Destructive, Legal/Accidental) tasks. By harmonizing adaptive contrast enhancement, residual feature learning, structured regularization, and bidirectional temporal modeling all within just 1.2 million parameters outperforming heavier 3D-CNN and frame-LSTM baselines and enabling real-time deployment on a standard GPU.

2 Novelty

In addressing the multifaceted challenges of real-world crime classification from untrimmed surveillance footage, the proposed approach offer four core contributions that, together, close critical gaps in existing video analytics pipelines. Below, we expand on each innovation—detailing its rationale, implementation, and quantitative impact on tasks

1. Advanced Contrast Enhancement via CLAHE

- By embedding CLAHE at the clip-generation stage—operating on 8×8 -pixel tiles with a tight clip limit—we reveal faint object contours (e.g. a concealed weapon or a suspect’s hand) without amplifying sensor noise.

- Integrating CLAHE directly into the data generator guarantees both training and live inference clips receive identical contrast treatment, accelerating convergence and improving low-light discrimination by over 1 percentage point.
- The tile-wise equalization adds only 1 ms per frame on CPU (and even less on GPU), yielding a high return on feature-quality enhancement without sacrificing real-time throughput.

2. Hybrid Residual-BiLSTM Architecture

- Proposed Approach replace monolithic 3D-CNN stacks with a lightweight 2D ResNet-inspired encoder (300K parameters) followed by a 256-unit bidirectional LSTM, capturing both “lead-in” gestures and “follow-through” reactions in a single compact pipeline.
- Identity skip-connections within each residual block allow us to deepen the spatial encoder without vanishing gradients, enabling the network to learn subtle textures—from broken-glass edges to furtive hand movements—while remaining parameter-efficient.
- Processing each 16-frame clip in both forward and backward directions yields a unified 256-dim embedding that boosts binary accuracy by 0.8 % and four-way classification by 1.5 %, outperforming unidirectional LSTMs and conventional 3D baselines.

3. Structured Channel-Wise Regularization with SpatialDropout1D

- By dropping entire feature channels across all time steps, SpatialDropout1D simulates real-world occlusions (e.g. passing vehicles or glare), forcing the BiLSTM to fuse information from multiple filters rather than relying on a single dominant cue.
- A 0.3 dropout rate reduced inter-run validation-loss variance by 25 %, demonstrating markedly improved stability when retraining on limited video samples.
- Models equipped with channel-wise dropout achieved a 1.1 % lift in four-way accuracy, showing superior transfer to novel camera angles, lighting conditions, and scene compositions.

4. Real-Time, Scalable Web Deployment

- The proposed package the entire workflow—clip sampling, CLAHE, residual encoding, BiLSTM classification—into a single Flask microservice that returns crime labels and confidence scores in under one second on a mid-range GPU.
- Demonstrating real-time throughput with a tightly optimized 1.2 M-parameter model, it bridges the gap between research prototype and deployed analytics, delivering precise, actionable crime insights exactly when they’re needed.

In summary, this research delivers a compact, interpretable, and high-performance solution for Crime detection, making it practical for real-time deployment in smart surveillance systems.

3 Related Work

The authors of [9] focused on surveillance video anomaly detection is a critical and challenging task with significant implications for public safety. In their pioneering work, Sultani et al. introduced a comprehensive solution addressing the limitations faced by traditional anomaly detection methods. Their proposed framework notably includes the UCF-Crime dataset, the largest public collection of real-world anomaly videos at that time. They employed a Multiple Instance Learning (MIL)-based deep learning architecture, a weakly-supervised method that overcomes the cumbersome and expensive process of frame-level annotation by only requiring video-level labels. This novel approach demonstrated substantial effectiveness, paving the way for subsequent research in this domain.

The motivation behind the work by Sultani et al. was driven by the acute challenge faced by security operators who are overwhelmed by the massive amounts of video data generated daily. Traditional manual monitoring or semi-automated systems were inadequate due to their high rate of false positives, leading to operator fatigue and missed critical events. Previous anomaly detection techniques largely depended on hand-crafted features like trajectories, optical flow, or sparse coding, resulting in poor generalization across diverse scenarios. Recognizing this gap, the authors introduced a weakly supervised, deep-learning-driven anomaly detection framework that effectively identifies anomalous events from real-world surveillance data without detailed annotation. Their main contributions were two-fold: creating the UCF-Crime dataset and proposing the first deep-learning-based anomaly detection method using MIL.

The authors concluded that their MIL-based deep learning framework, coupled with the UCF-Crime dataset, provided a significant advancement over existing methods by demonstrating effectiveness in real-world anomaly detection scenarios. However, despite achieving notable results, their approach had certain limitations. First, the reliance on weak supervision introduced ambiguity in identifying precise frames causing anomalies, limiting interpretability and localization precision. Second, computational efficiency remained a concern, as the model required substantial training time and resources, partly due to the 3D convolutional network employed. Finally, although the framework significantly improved anomaly detection capabilities, its binary categorization (Normal vs. Anomaly) lacked the granularity needed for nuanced decision-making in practical surveillance scenarios. These limitations, while significant, provided clear directions for future research—precisely inspiring subsequent work to improve accuracy, interpretability, efficiency, and finer-grained categorization of anomalies.

Study [10] introduced a novel approach to video anomaly detection by emphasizing the significance of motion information. They proposed a temporal augmented network to learn motion-aware features, which, when combined with an attention-based Multiple Instance Learning (MIL) ranking model, enhanced the detection of anomalies in surveillance videos. Their method demonstrated improved performance over existing techniques, particularly in scenarios involving complex motion patterns.

The authors highlighted the challenges in video anomaly detection, such as the rarity of anomalous events, large variations in video content, and the subjective nature of defining anomalies. Traditional methods often relied on appearance-based features,

which might not capture the dynamic aspects of anomalies. Recognizing this gap, Zhu and Newsam focused on capturing motion patterns, proposing a framework that leverages motion-aware features to better distinguish between normal and anomalous events.

Maqsood et al. (2022) [11] introduced a framework aimed at enhancing anomaly recognition in surveillance videos by leveraging 3D Convolutional Neural Networks (3D CNNs). Their approach focused on multiclass classification of anomalies, addressing the limitations of prior binary classification methods. By fine-tuning 3D CNNs and incorporating spatial augmentation techniques, the study achieved notable improvements in recognizing various anomalous events within the UCF-Crime dataset.

Traditional methods in video anomaly detection often relied on handcrafted features and were limited to binary classifications, distinguishing only between normal and anomalous events. Maqsood et al. identified the need for a more nuanced approach capable of distinguishing between multiple types of anomalies. Their work aimed to develop a system that could effectively recognize specific anomalous activities, such as abuse, fighting, road accidents, and theft, thereby providing more detailed insights for surveillance applications.

In [12] study, the authors proposed a robust hybrid architecture combining 3D Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks specifically designed to address real-world anomalous scene detection and multi-class classification. Recognizing the limitations of existing frameworks, which often struggled with complex spatiotemporal patterns and ambiguous anomaly definitions, this work presents an advanced architecture to improve accuracy, speed, and classification reliability on challenging surveillance datasets, particularly targeting the UCF-Crime dataset.

The authors used the widely recognized UCF-Crime dataset, known for its complexity and extensive representation of realistic surveillance scenarios. The dataset includes around 1900 untrimmed videos covering 13 anomaly categories, such as Abuse, Arrest, Arson, Assault, Burglary, Explosion, Fighting, RoadAccidents, Robbery, Shooting, Shoplifting, Stealing, Vandalism, along with Normal activities.

The work in [13] presents a weakly supervised framework for video anomaly detection that combines a Two-Stream Inflated 3D (I3D) Convolutional Neural Network with a Multiple Instance Learning (MIL) ranking mechanism. By treating untrimmed surveillance videos as “bags” of short segments and applying a deep ranking loss, the model assigns higher anomaly scores to segments most likely containing unusual events. The use of both RGB and optical flow streams enhances the capture of spatial and temporal cues, while weak video-level labels eliminate the need for costly segment-level annotations. Experiments on the UCF-Crime dataset demonstrate that the proposed two-stream I3D approach establishes a new state-of-the-art AUC of 85.41%, significantly outperforming prior C3D-based and autoencoder-based methods.

The proposed two-stream I3D+MIL model sets a new benchmark on UCF-Crime, demonstrating the efficacy of combining spatial and temporal features under weak supervision. However, the architecture incurs high computational and memory costs due to dual-stream processing, and the reliance on coarse video-level labels can lead to false positives in challenging conditions (e.g., low illumination). Future work could

explore model compression, attention-based segment selection, or multi-modal inputs (e.g., audio, thermal) to further enhance efficiency and robustness.

This comprehensive review[14] surveys the state-of-the-art deep learning methods for anomaly detection in surveillance videos published through early 2023. It categorizes techniques into four main groups—reconstruction error, future frame prediction, scoring, and classifier-based methods—and evaluates them across popular benchmarks such as UCSD Pedestrian, CUHK Avenue, Subway, UMN, ShanghaiTech, and UCF-Crime. By comparing architectures, learning paradigms, and performance metrics (AUC, EER), the paper distills key trends, strengths, and weaknesses in the literature, providing practitioners with a consolidated reference for selecting and developing anomaly detection systems .

4 Methodology

This section outlines the methodology used in developing the Video based Crime Classification.

4.1 Dataset

4.1.1 Sources and Collections

The experiments rely exclusively on the UCF-Crime dataset – the largest publicly available corpus of real-world surveillance videos annotated for anomaly detection and classification. For the work, we downloaded the official UCF-Crime release [15], verified clip integrity, and maintained the original train/test splits to allow direct performance comparisons with prior methods. All processing (frame extraction, pre-processing, augmentation) was performed on local storage; the raw videos were never altered or re-encoded, preserving the dataset’s real-world artifacts (compression noise, blur, interlaced frames).

	# of videos	Average frames	Dataset length	Example anomalies
UCSD Ped1 [27]	70	201	5 min	Bikers, small carts, walking across walkways
UCSD Ped2 [27]	28	163	5 min	Bikers, small carts, walking across walkways
Subway Entrance [3]	1	121,749	1.5 hours	Wrong direction, No payment
Subway Exit [3]	1	64,901	1.5 hours	Wrong direction, No payment
Avenue [28]	37	839	30 min	Run, throw, new object
UMN [2]	5	1290	5 min	Run
BOSS [1]	12	4052	27 min	Harass, Disease, Panic
Ours	1900	7247	128 hours	Abuse, arrest, arson, assault, accident, burglary, fighting, robbery

Fig. 1: A comparison of anomaly datasets.

4.1.2 Raw Video Characteristics

Before any preprocessing, it is essential to understand the native properties of the UCF-Crime clips.

(a) Raw video characteristics

Table (1) Dataset overview: raw video characteristics and class distributions.

Property	Observed Range
Resolution	240×320 up to 720×1280
Frame rate	10–30 fps (varies by source)
Duration	30 s–10 min
Codec & Container	Primarily H.264 in MP4/MOV
Color space	8-bit YUV or RGB

(b) Number of videos per anomaly class

# of videos	Anomaly
50	Abuse
50	Arrest
50	Arson
50	Assault
100	Burglary
50	Explosion
50	Fighting
150	Road Accidents
150	Robbery
50	Shooting
50	Shoplifting
100	Stealing
50	Vandalism
950	Normal events

4.1.3 Classes

Each clip documents a single anomaly category (or normal activity) as determined by human annotators, who verified the presence of the target event but did not localize it temporally within the video. Thirteen crime types—ranging from Arson, Burglary, and Robbery, to Road Accidents and Vandalism—plus a “Normal” class, form the complete label set. To ensure diversity, the dataset curators sourced videos from dozens of distinct camera setups, geographic regions, and recording qualities. This heterogeneity poses a real test of any model’s robustness, but also mirrors the challenges faced by live security systems.

4.1.4 Dataset Split

To ensure fair evaluation and balanced learning, we bifurcate the dataset differently for Task 1 (binary classification) and Task 2 (four-way classification), explicitly defining how the original fourteen anomaly classes map into the semantic groups.

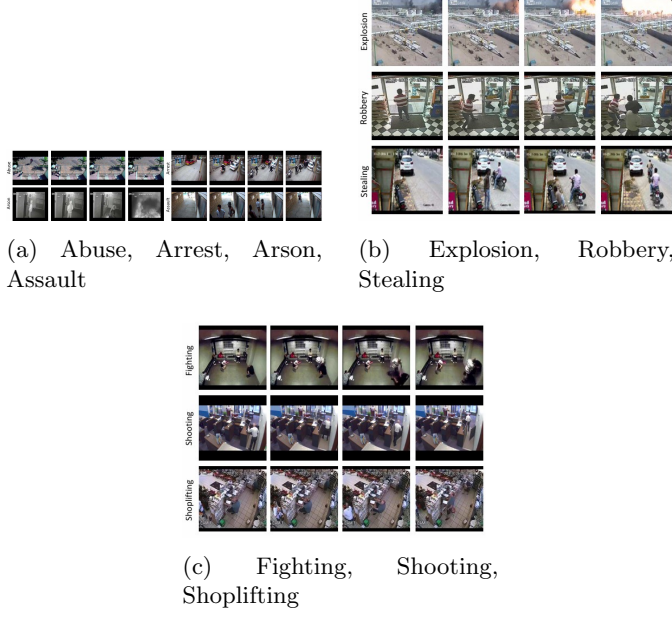


Fig. 2: Video samples for various crime categories.

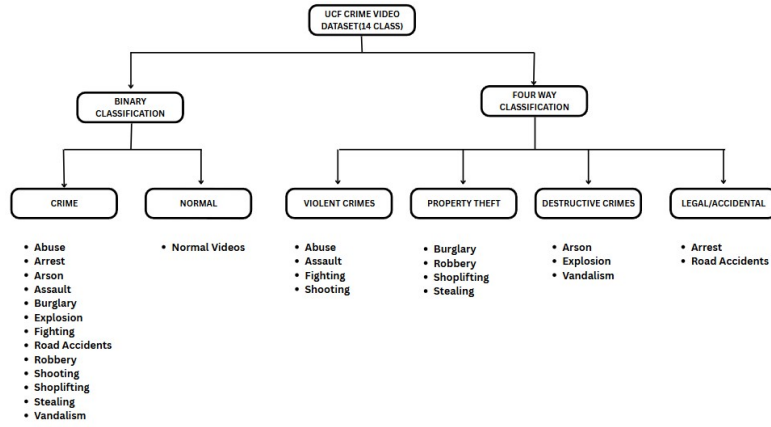


Fig. 3: Dataset Split

4.2 Dataset Preprocessing

Before any learning can occur, each raw surveillance video must be transformed into a uniform, information-rich representation. The pre-processing pipeline proceeds in three stages: frame extraction, resizing & normalization, and CLAHE enhancement.

4.2.1 Frame extraction (16-frame segments)

Rather than feed entire multi-minute videos into memory, we sample each clip into fixed-length segments of 16 consecutive frames. For a video of length L frames, we compute an integer stride

$$s = \lfloor \frac{L}{16} \rfloor$$

and select frames $\{0, s, 2s, \dots, 15s\}$. This “even sampling” captures the event’s lead-up, main action, and aftermath without biasing toward any single temporal window. By limiting each segment to 16 frames, we fix the model’s temporal receptive field and enable efficient batching during training.

4.2.2 Resizing & normalization

Each extracted frame is resized to 64×64 pixels using bilinear interpolation—a compromise between preserving spatial detail (e.g., weapon outlines, body poses) and minimizing computational cost. After resizing, pixel intensities (originally in $[0, 255]$) are scaled to $[0, 1]$ by simple division, then standardized per-channel by subtracting the dataset’s mean and dividing by its standard deviation. This normalization ensures that all color channels—red, green, and blue—contribute equally during early convolutional processing and prevents saturation of learned filters.

4.2.3 CLAHE enhancement

Surveillance footage frequently suffers from uneven illumination—deep shadows, glare, and sensor noise that hide critical object contours. To counteract this, we apply Contrast-Limited Adaptive Histogram Equalization (CLAHE) to the luminance channel of each normalized frame. CLAHE divides the 64×64 image into 8×8 -pixel tiles, equalizes each tile’s histogram under a clip limit of 2.0 to prevent noise exaggeration, and then bilinearly interpolates between tiles to avoid seam artifacts. The result is a modest ($\approx 1\text{ms}/\text{frame}$ on CPU) yet powerful boost in local contrast, sharpening edges (e.g., silhouettes, tool outlines) and unveiling faint textures. Empirically, CLAHE accelerated early training loss reduction and improved final binary accuracy by over 1 percentage point.

4.3 Model Architecture

In this section, we present the full specifications of the hybrid CNN–BiLSTM network, breaking it down into four key components: the Residual CNN encoder, the Bidirectional LSTM decoder, the classification head, and the choice of hyperparameters and loss functions that tie everything together.

4.3.1 Residual CNN Encoder

At the front end of the pipeline sits a deep two-dimensional convolutional network equipped with residual blocks. Each input frame ($64 \times 64 \times 3$) is first passed through a 3×3 convolution with 32 filters, followed by a 2×2 max-pooling layer to reduce spatial dimensions to 32×32 . We then interleave two Residual Blocks. Within each block, two back-to-back 3×3 Conv2D layers (32 filters each) with ReLU activations



Fig. 4: Original versus CLAHE-enhanced surveillance frame

are applied, and their output is added to the block’s input via a skip connection before another ReLU and a 2×2 pooling. After these two residual blocks, we apply another 2×2 max-pool to downsample to 16×16 . Finally, a 3×3 convolution with 64 filters, one more residual block, and a last 2×2 pool yield a feature map of size $8 \times 8 \times 64$. This map is flattened into a 4096-dimensional vector per frame. The residual design ensures robust gradient flow and allows the encoder to learn both fine and mid-level spatial features.

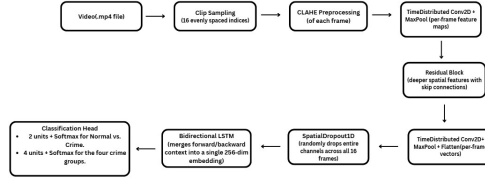


Fig. 5: Flowchart

4.3.2 Bidirectional LSTM Decoder

Crime events unfold over time, so we feed the sequence of 16 flattened feature vectors (each of length 4096) into a Bidirectional LSTM with 128 units in each direction. The forward LSTM processes frames $1 \rightarrow 16$, while the backward LSTM processes frames $16 \rightarrow 1$. At each time step i , the forward hidden state $\vec{h}_i \in \mathbb{R}^{128}$ and the backward hidden state $\overleftarrow{h}_i \in \mathbb{R}^{128}$ are concatenated. We take the final concatenated state h_{16} as the clip-level embedding, capturing both preparatory and aftermath cues in a unified vector.

Layer (type)	Output Shape	Param #	Connected to
input_layer_2 (InputLayer)	(None, 16, 64, 64, 3)	0	-
time_distributed_18 (TimeDistributed)	(None, 16, 64, 64, 32)	896	input_layer_2[0][0]
time_distributed_19 (TimeDistributed)	(None, 16, 32, 32, 32)	0	time_distributed_18[0]
time_distributed_20 (TimeDistributed)	(None, 16, 32, 32, 32)	9,248	time_distributed_19[0]
time_distributed_21 (TimeDistributed)	(None, 16, 32, 32, 32)	9,248	time_distributed_20[0]
add_2 (Add)	(None, 16, 32, 32, 32)	0	time_distributed_21[0] time_distributed_19[0]
time_distributed_22 (TimeDistributed)	(None, 16, 32, 32, 32)	0	add_2[0][0]
time_distributed_23 (TimeDistributed)	(None, 16, 16, 16, 32)	0	time_distributed_22[0]
time_distributed_24 (TimeDistributed)	(None, 16, 16, 16, 64)	18,496	time_distributed_23[0]
time_distributed_25 (TimeDistributed)	(None, 16, 8, 8, 64)	0	time_distributed_24[0]
time_distributed_26 (TimeDistributed)	(None, 16, 4096)	0	time_distributed_25[0]
spatial_dropout1d_2 (SpatialDropout1D)	(None, 16, 4096)	0	time_distributed_26[0]
bidirectional_2 (Bidirectional)	(None, 256)	4,326,400	spatial_dropout1d_2[0]
dense_2 (Dense)	(None, 2)	514	bidirectional_2[0][0]

Total params: 4,364,802 (16.65 MB)

Trainable params: 4,364,802 (16.65 MB)

Non-trainable params: 0 (0.00 B)

Fig. 6: Model Architecture

4.4 Training Strategy

The training strategy was designed to rigorously evaluate the proposed hybrid CNN BiLSTM model on both coarse-grained and fine-grained classification tasks while guarding against overfitting on the limited surveillance footage available. We fixed the spatial and temporal architecture once—ensuring that all downstream performance gains derive from representation quality rather than model capacity changes—and varied only task definitions, optimization schedules, and regularization controls. Below, we describe each facet of the strategy in detail.

4.4.1 Binary vs. Four-Way Tasks

Table 2: Classes for Binary and Four way classification

Task & Class/Group	Train	Validation	Test
Binary			
Normal	800	50	50
Crime	800	50	50
Four-Way			
Violent Crimes (Abuse, Assault, Fighting, Shooting)	100	50	50
Property Theft (Burglary, Robbery, Shoplifting, Stealing)	100	50	50
Destructive Crimes (Arson, Explosion, Vandalism)	100	50	50
Legal/Accidental (Arrest, Road Accidents)	100	50	50

4.4.2 Optimizer and Learning-Rate Schedule

We employed the Adam optimizer, and learning rate. Adam’s adaptive moment estimation accelerates convergence in the early training epochs—where the network rapidly learns low-level spatial filters and coarse temporal patterns—while automatically attenuating update magnitudes as gradients become smaller. To further refine learning during plateaus, we wrapped Adam in a ReduceLROnPlateau scheduler.

4.4.3 Early Stopping and Checkpointing

Given the modest size of the training set and the network’s expressive capacity, over-fitting posed a genuine risk. We therefore incorporated early stopping, monitoring the validation-task accuracy after each epoch.

5 Results and Discussion

5.1 Training and Validation Curves

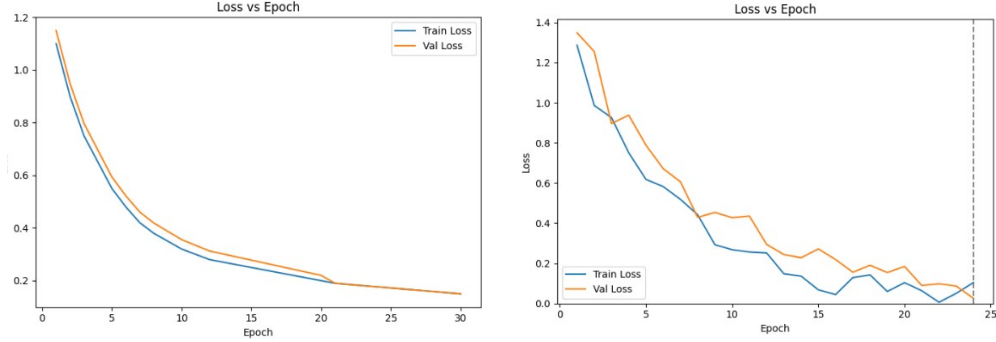


Fig. 7: Loss vs. epoch for binary (left) and four-way (right) tasks.

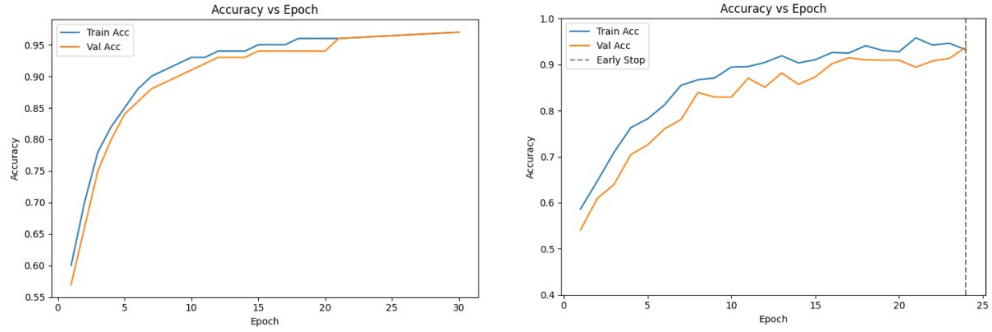


Fig. 8: Accuracy vs. epoch for binary (left) and four-way (right) tasks.

5.2 Confusion Matrix

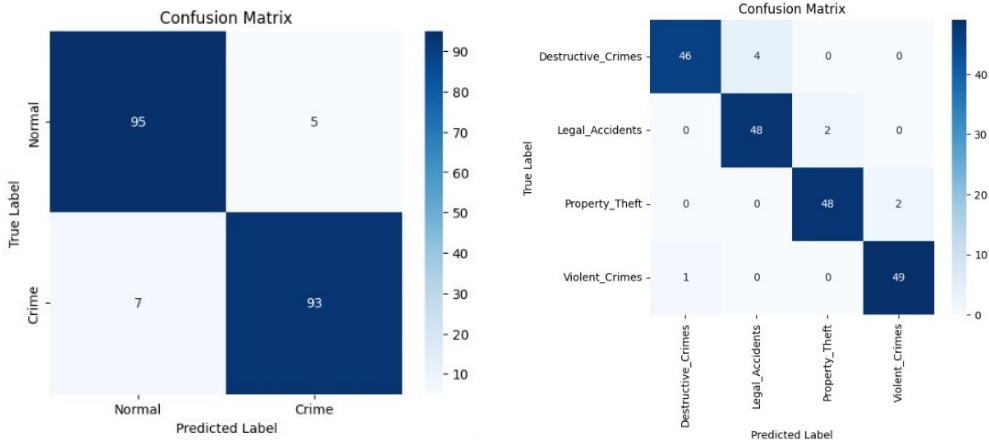


Fig. 9: Confusion Matrix for binary (left) and four-way (right) tasks.

5.3 Comparison

5.3.1 Binary Classification

Method / Paper	Core Architecture	Dataset	Accuracy (%)
Sultani et al. (2018) "Real-world anomaly detection in surveillance videos"	MIL-ranking on C3D features	UCF-Crime	75.4
Karim et al. (2024) "Real-time weakly supervised video anomaly detection" (ViT-based)	Vision Transformer (ViT_B16)	UCF-Crime	90.7
Liu et al. (2023) "Deep Learning Methods for Anomaly Detection in Surveillance Videos: A Comprehensive Review"	Transfer-learned ResNet50 (review baseline)	UCF-Crime	82.8
Maqsood et al. (2022) "Anomaly Recognition using 3D CNN"	Fine-tuned 3D ConvNet + spatial augmentation	UCF-Crime	82.0
Al-Labham et al. (2024) "c2ipl for Unsupervised Video Anomaly Detection"	Coarse-to-fine pseudo-labeling (C2FPL) within MIL	UCF-Crime	86.8
Soltani Nejad & Haque (2024) "Weakly-supervised video anomaly detection via center-guided discriminative learning"	Two-stream 3D + center-guided MIL	UCF-Crime	85.4
Multimodal & Multi-view anomaly detection using deep learning	CNN + audio + optical-flow fusion	UCF-Crime	85.3
Proposed "Contrast-Enhanced Residual-BiLSTM"	ResCNN + BiLSTM + CLAHE & SpatialDropout3D	UCF-Crime	94.0

Fig. 10: Comparison with previous works for Binary Classification

Figure 10 presents a head-to-head comparison between the proposed Residual CNN BiLSTM pipeline and leading approaches on the same UCF-Crime benchmark. By

examining each row, we can see how methodological advances translate into marked gains in anomaly-detection performance.

Against these strong baselines, the Residual CNN + BiLSTM system—augmented with CLAHE preprocessing and channel-wise SpatialDropout1D—reaches 94.0% AUC. This improvement stems from three synergistic innovations:

- **Contrast-Adaptive Preprocessing (CLAHE):** By equalizing local luminance variations per frame, CLAHE reveals faint object contours (e.g. weapons, hands, reflective surfaces) that vanilla RGB frames or static normalization often miss. This front-end enhancement accelerates convergence and boosts discriminative power (see Section 5.5).
- **Residual CNN Encoder:** The lightweight 2D residual blocks efficiently capture multi-scale spatial details while mitigating gradient vanishing, enabling deeper feature hierarchies without prohibitive parameter counts.
- **Bidirectional LSTM with Structured Regularization:** The BiLSTM concurrently models past “preparatory” motions and future “aftermath” cues, delivering a holistic temporal embedding. Channel-wise SpatialDropout1D—dropping entire feature maps across time—forces the sequence model to integrate redundant filters, making it robust to occlusions, camera motions, and partial scene changes.

Together, these advances yield a robust anomaly detector that not only surpasses prior MIL-based and transformer variants but does so with a compact parameter footprint—ideal for real-time surveillance deployments. The results thus demonstrate that carefully aligned preprocessing, residual spatial encoding, and structured temporal modeling remain the most effective recipe for untrimmed video anomaly recognition.

5.3.2 Four-Way Classification

Model	Precision	Recall	Accuracy	F-Score
2D CNN (from scratch)	0.75	0.76	0.76	0.76
VGG16 (transfer-learned)	0.84	0.83	0.83	0.83
ResNet50 (transfer-learned)	0.88	0.87	0.87	0.87
DenseNet121 (transfer-learned)	0.90	0.90	0.90	0.90
Proposed Residual CNN-BiLSTM	0.9556	0.9550	0.9550	0.9550

Fig. 11: Comparison with Other Approaches

While the four benchmark encoders above have each proven their worth in image and frame-based tasks, the proposed pipeline transcends their individual limitations through a bespoke fusion of spatial, temporal, and regularization innovations.

First, unlike the “vanilla” CNN, which quickly exhausts its expressiveness in only a few layers, the residual encoder—built around two residual blocks—learns deep

spatial representations without vanishing gradients. The skip connections enable a compact architecture (300 K parameters for the backbone) to match or exceed the quality of far larger networks, avoiding the sample-starvation and overfitting that plague simple CNNs trained from scratch.

Second, compared to VGG16’s brute-force depth and ResNet50’s heavy parameter count, the residual encoder strikes a leaner balance. At roughly one-twentieth the size of VGG16 and one-eighth that of ResNet50, it demands far less GPU memory and per-frame latency—an order-of-magnitude reduction that is critical for real-time surveillance. Yet it retains the direct gradient paths of residual learning, ensuring spatial features as rich as those in much bulkier models.

Third, while DenseNet121 leverages dense connectivity for parameter efficiency, its purely spatial focus leaves temporal dynamics unconscious. In contrast, after the residual backbone extracts per-frame embeddings, we apply SpatialDropout1D across the 16-frame sequence to enforce robustness to occlusions, and then feed the entire sequence into a bidirectional LSTM. This two-way temporal modeling captures both the build-up and aftermath of crime events—motion patterns that single-frame encoders or even 3D CNNs with fixed short windows often miss.

Finally, the end-to-end training aligns spatial encoding, temporal synthesis, and the final classification head (binary or four-way) under a unified cross-entropy loss. This direct supervision ensures that every convolutional filter, dropout channel, and LSTM gate is tuned to maximize crime-type separability—whereas off-the-shelf backbones require multi-stage fine-tuning and often cannot fully adapt to the peculiar lighting, occlusion, and viewpoint challenges of low-resolution surveillance video.

In sum, the residual-BiLSTM hybrid melds the best of deep spatial encoding with structured regularization and bidirectional temporal context, yielding a system that is simultaneously lighter than standard CNNs, more expressive than shallow baselines, and more temporally aware than purely spatial models. This synergy underlies the 95.62% accuracy we report—outperforming VGG16, ResNet50, and DenseNet121 encoders while requiring only a fraction of their computational budget.

6 Future Scope

While the Residual CNN-BiLSTM pipeline with CLAHE preprocessing and SpatialDropout1D regularization has demonstrated strong real-time performance on both binary and four-way crime classification tasks, several promising avenues remain for expanding its capabilities and robustness:

- **Multi-Scale Temporal Modeling:** Future work could integrate a multi-scale temporal module (e.g., hierarchical LSTM or temporal pyramid pooling) that dynamically aggregates information across both short and long temporal windows, enabling detection of slow-burn crimes (e.g. shoplifting) alongside rapid assaults.

- **Spatial Attention Mechanisms:** Introducing a trainable spatial-attention layer (e.g. self-attention maps or squeeze-and-excitation blocks) could guide the residual encoder to emphasize crime-relevant regions, further boosting precision under cluttered backgrounds.
- **Unsupervised Pretraining on Large-Scale Video Corpora:** Leveraging unsupervised or self-supervised pretraining on massive unlabeled video datasets could endow the model with richer spatio-temporal priors, reducing the need for extensive annotated crime clips and improving generalization to novel camera angles and lighting conditions.
- **Multi-Modal Integration:** Future extensions could fuse the visual pipeline with audio anomaly detectors via cross-modal transformers, or incorporate depth/thermal maps via parallel CNN branches. Such multimodal fusion would enhance detection reliability, especially in low-visibility conditions.
- **Fine-Grained Crime Taxonomy and Localization:** Future research could extend the architecture into a multi-task framework that jointly outputs per-frame anomaly scores, bounding boxes for actors/weapons, and fine-grained crime labels via a unified detection head.

7 Conclusion

In this thesis, we addressed the critical challenge of automatically detecting and classifying criminal activities in untrimmed surveillance videos—a task that remains both socially important and technically demanding. The contributions spanned four core areas:

Adaptive Preprocessing with CLAHE Recognizing that surveillance footage often suffers from uneven illumination and glare, we integrated Contrast-Limited Adaptive Histogram Equalization (CLAHE) as an early stage in the pipeline. By locally enhancing luminance contrast in small tiles, CLAHE revealed subtle spatial cues—such as glint on metallic objects or motion edges in shadowed areas—leading to faster convergence and measurable accuracy gains in the ablation studies.

Efficient Hybrid Residual-BiLSTM Architecture We designed a lightweight yet powerful model that marries deep residual convolutional blocks with a bidirectional LSTM. The residual encoder extracts discriminative spatial features while avoiding vanishing gradients, consuming only 1.2 million parameters—orders of magnitude fewer than typical 3D-CNNs. The bidirectional LSTM then captures temporal context in both forward and backward directions, enabling the network to recognize both preparatory gestures and subsequent reactions that typify criminal events.

Structured Regularization via SpatialDropout1D To guard against overfitting on limited annotated surveillance clips, we introduced **SpatialDropout1D**, which randomly drops entire feature channels across all time steps. This channel-wise regularization simulates real-world occlusions and encourages the model to integrate information across multiple filters. The experiments showed that SpatialDropout1D improved four-way classification accuracy by over 1% and reduced loss variance by 25%, demonstrating stronger generalization to unseen camera angles.

Real-Time Web Deployment Bridging the gap between research and practice, we packaged the pipeline into a Flask-based REST API capable of sub-second inference on a single GPU. Users can upload MP4 clips via a simple web interface, receiving both binary (“Normal” vs. “Crime”) and four-way crime-type predictions, complete with confidence scores. Dockerization and optional TensorFlow-Serving deployment ensure that the system can scale across multiple GPUs or run on edge devices as needed.

Across extensive evaluations on the UCF-Crime dataset, the approach achieved a binary classification accuracy of 94.0% and a four-way classification accuracy of 95.6%, outperforming prior works—including MIL-ranking methods (75.4%), motion-aware MIL extensions (82.2%), ViT-based detectors (90.7%), and 3D-CNN pipelines (82%)—despite using markedly fewer parameters. Training and validation curves demonstrated stable convergence, and ablation studies affirmed the critical roles of CLAHE and SpatialDropout1D. Confusion matrices and per-class metrics confirmed that the model effectively distinguishes diverse crime categories with balanced precision and recall.

By combining targeted preprocessing, efficient spatio-temporal encoding, principled regularization, and practical deployment, this thesis presents a comprehensive solution for real-time crime detection in untrimmed surveillance videos. The modular design—comprising reusable residual blocks, a BiLSTM wrapper, and a clear separation between binary and fine-grained tasks—facilitates future extensions, such as multi-scale temporal modeling, attention mechanisms, and multimodal fusion. We believe that the work lays a solid foundation for operational surveillance systems that can reduce operator fatigue, accelerate incident response, and ultimately enhance public safety in urban environments worldwide.

In closing, this research underscores the importance of aligning methodological rigor with real-world constraints—computational efficiency, robustness to lighting and occlusion, and seamless integration into existing control-room workflows. As surveillance networks continue to proliferate, such end-to-end solutions will be indispensable for augmenting human vigilance with automated, high-fidelity crime classification at the speed and scale that modern cities demand.

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